Assignment 3: Data Exploration

Joy Wu

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OVERVIEW

This exercise accompanies the lessons in Environmental Data Analytics on Data Exploration.

Directions

1. Rename this file `<FirstLast>_A03_DataExploration.Rmd` (replacing `<FirstLast>` with your first and last name).
2. Change "Student Name" on line 3 (above) with your name.
3. Work through the steps, **creating code and output** that fulfill each instruction.
4. Assign a useful **name to each code chunk** and include ample **comments** with your code.
5. Be sure to **answer the questions** in this assignment document.
6. When you have completed the assignment, **Knit** the text and code into a single PDF file.
7. After Knitting, submit the completed exercise (PDF file) to the dropbox in Canvas.

TIP: If your code extends past the page when knit, tidy your code by manually inserting line breaks.

TIP: If your code fails to knit, check that no `install.packages()` or `View()` commands exist in your code.

Set up your R session

1. Load necessary packages (tidyverse, lubridate, here), check your current working directory and upload two datasets: the ECOTOX neonicotinoid dataset (ECOTOX_Neonicotinoids_Insects_raw.csv) and the Niwot Ridge NEON dataset for litter and woody debris (NEON_NIWO_Litter_massdata_2018-08_raw.csv). Name these datasets "Neonics" and "Litter", respectively. Be sure to include the subcommand to read strings in as factors.

```
#Load necessary packages
library(tidyverse)
library(lubridate)
library(here)

#Check current working directory
getwd()
```

```
## [1] "/home/guest/EDE_Fall2025"
```

```r
#Load datasets
Neonics <- read.csv(
  file = here('Data/Raw/ECOTOX_Neonicotinoids_Insects_raw.csv'),
  stringsAsFactors = TRUE,
)
Litter <- read.csv(
  file = here('Data/Raw/NEON_NIWO_Litter_massdata_2018-08_raw.csv'),
  stringsAsFactors = TRUE,
)
```

Learn about your system

2. The neonicotinoid dataset was collected from the Environmental Protection Agency's ECOTOX Knowledgebase, a database for ecotoxicology research. Neonicotinoids are a class of insecticides used widely in agriculture. The dataset that has been pulled includes all studies published on insects. Why might we be interested in the ecotoxicology of neonicotinoids on insects? Feel free to do a brief internet search if you feel you need more background information.

   Answer: These datasets can help us to understand the relationship between chemical usage and the effects they have. They can also help us to conduct sensitivity analysis among species and to create policies and risk assessments using the results.

3. The Niwot Ridge litter and woody debris dataset was collected from the National Ecological Observatory Network, which collectively includes 81 aquatic and terrestrial sites across 20 ecoclimatic domains. 32 of these sites sample forest litter and woody debris, and we will focus on the Niwot Ridge long-term ecological research (LTER) station in Colorado. Why might we be interested in studying litter and woody debris that falls to the ground in forests? Feel free to do a brief internet search if you feel you need more background information.

   Answer: We are interested in studying these because the data helps us to understand the decomposition process and its ecological impacts. From there we can conduct more research on the broader ecology systems and how each individual component in the system affect the broader system.

4. How is litter and woody debris sampled as part of the NEON network? Read the NEON_Litterfall_UserGuide.pdf document to learn more. List three pieces of salient information about the sampling methods here:

   Answer: 1.In sites with forested tower airsheds, the litter sampling is targeted to take place in 20 40mx40mplots. In sites with low-statured vegeta on over the tower airsheds, litter sampling is targeted to take place in 4 40m x 40m tower plots (to accommodate co-located soil sampling) plus 26 20m x 20m plots. 2.litter is defined as material that is dropped from the forest canopy and has a bu end diameter <2cm andalength <50 cm. It is collected in elevated 0.5m^2 PVC traps. 3. Ground traps are sampled once per year. Target sampling frequency for elevated traps varies by vegeta on present at the site, with frequent sampling (1x every 2weeks) in deciduous forest sites during senescence, and infrequent year-round sampling (1x every 1-2 months) at evergreen sites.

Obtain basic summaries of your data (Neonics)

5. What are the dimensions of the dataset?

```r
dim(Neonics)
```

```
## [1] 4623   30
```

6. Using the `summary` function on the "Effect" column, determine the most common effects that are studied. Why might these effects specifically be of interest? [Tip: The `sort()` command is useful for listing the values in order of magnitude…]

```r
effect_summary <- summary(Neonics$Effect)
sorted_effects <- sort(effect_summary, decreasing = TRUE) #sorting effects from largest to smallest
print(sorted_effects)
```

```
##       Population        Mortality         Behavior Feeding behavior 
##             1803             1493              360              255 
##     Reproduction      Development        Avoidance         Genetics 
##              197              136              102               82 
##        Enzyme(s)           Growth       Morphology    Immunological 
##               62               38               22               16 
##     Accumulation     Intoxication     Biochemistry          Cell(s) 
##               12               12               11                9 
##       Physiology        Histology       Hormone(s) 
##                7                5                1
```

Answer: We are interested in studying the effects because they show the impacts of insecticides on insect populations.

7. Using the `summary` function, determine the six most commonly studied species in the dataset (common name). What do these species have in common, and why might they be of interest over other insects? Feel free to do a brief internet search for more information if needed.[TIP: Explore the help on the `summary()` function, in particular the `maxsum` argument…]

```r
species_counts <- summary(Neonics$Species.Common.Name, maxsum = Inf)
species_sorted <- sort(species_counts, decreasing = TRUE)
top6 <- head(species_sorted, 6)
print(top6)
```

```
##            Honey Bee        Parasitic Wasp Buff Tailed Bumblebee 
##                  667                   285                   183 
##   Carniolan Honey Bee           Bumble Bee     Italian Honeybee 
##                  152                   140                   113
```

Answer: The top six species are all bees. They might be of interest over other insects because they are highly exposed of insectcides than other species due to their activity patterns. They contribute to the whole ecosystem working as pollinators, and they might be more sensitive to insectcides being used.

8. Concentrations are always a numeric value. What is the class of `Conc.1..Author.` column in the dataset, and why is it not numeric? [Tip: Viewing the dataframe may be helpful…]

```r
class(Neonics$Conc.1..Author.)
```

```
## [1] "factor"
```

```r
concentrations <- summary(Neonics$Conc.1..Author.)
print(head(concentrations,20))
```

```
##  0.37/    10/    NR/     NR      1   1023  0.40/     2/     10 0.053/    100
##    208    127    108     94     82     80     69     63     62     59     56
##    50/   0.5/   0.03  0.05/   0.45   0.1/  0.45/   1.0/  2.27/
##     51     45     44     43     43     42     40     40     40
```

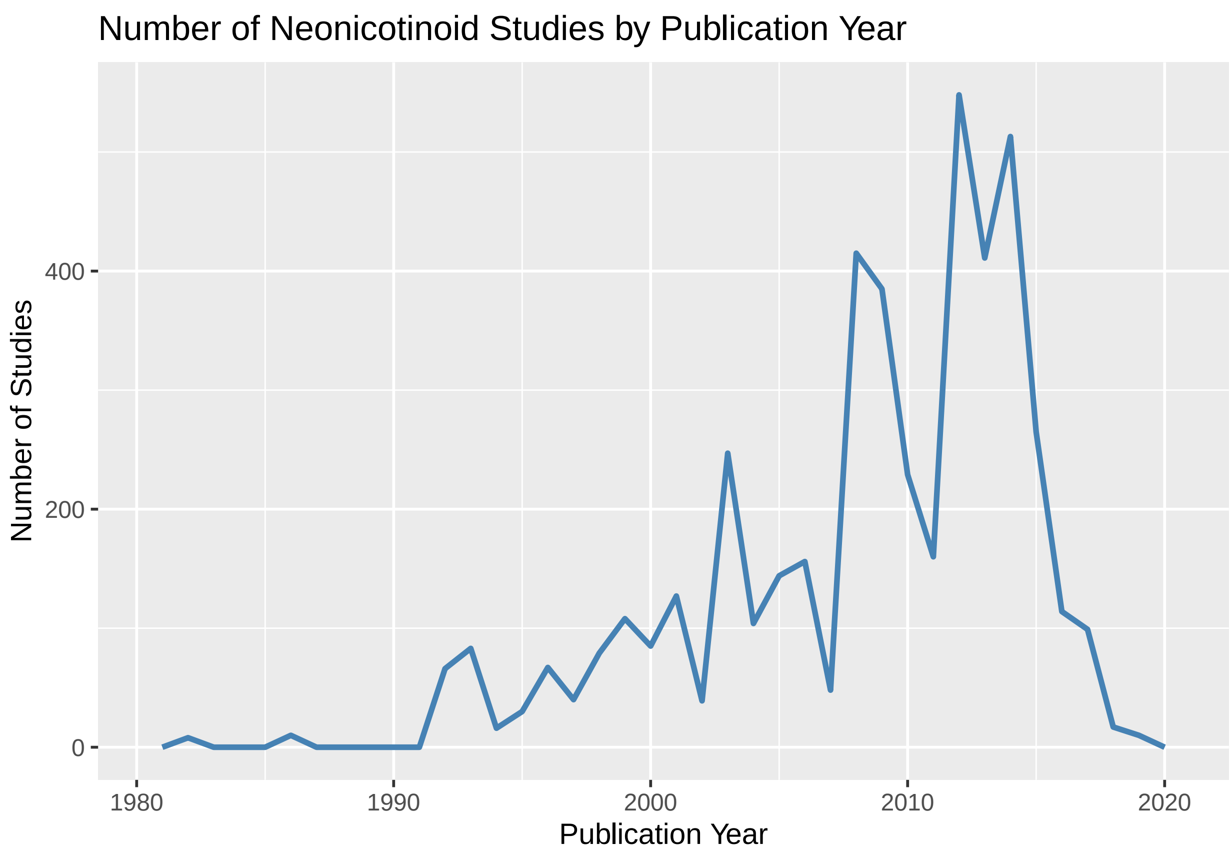
> Answer: It is not numeric here because the values that are stored with this name have mixed formating and non-mueric values are present as well.

Explore your data graphically (Neonics)

9. Using `geom_freqpoly`, generate a plot of the number of studies conducted by publication year.

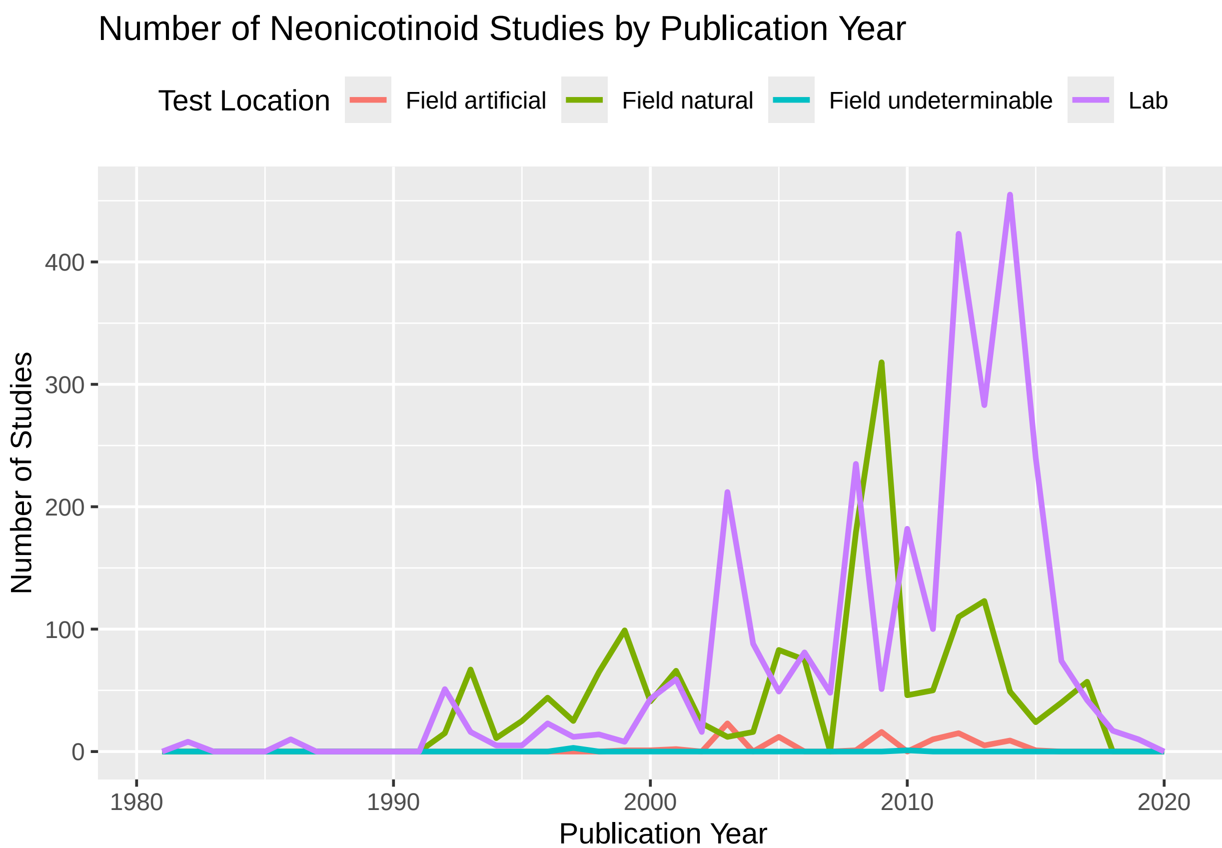
```r
ggplot(Neonics, aes(x = Publication.Year)) +
  geom_freqpoly(binwidth = 1, color = "steelblue", size = 1) +
  labs(
    title = "Number of Neonicotinoid Studies by Publication Year",
    x = "Publication Year",
    y = "Number of Studies"
  ) +
  theme(legend.position = "top")
```

```
## Warning: Using ‘size‘ aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use ‘linewidth‘ instead.
## This warning is displayed once every 8 hours.
## Call ‘lifecycle::last_lifecycle_warnings()‘ to see where this warning was
## generated.
```



10. Reproduce the same graph but now add a color aesthetic so that different Test.Location are displayed as different colors.

```
ggplot(Neonics, aes(x = Publication.Year, color = Test.Location)) +
  geom_freqpoly(binwidth = 1, size = 1) +
  labs(
    title = "Number of Neonicotinoid Studies by Publication Year",
    x = "Publication Year",
    y = "Number of Studies",
    color = "Test Location"
  ) +
  theme(legend.position = "top")
```



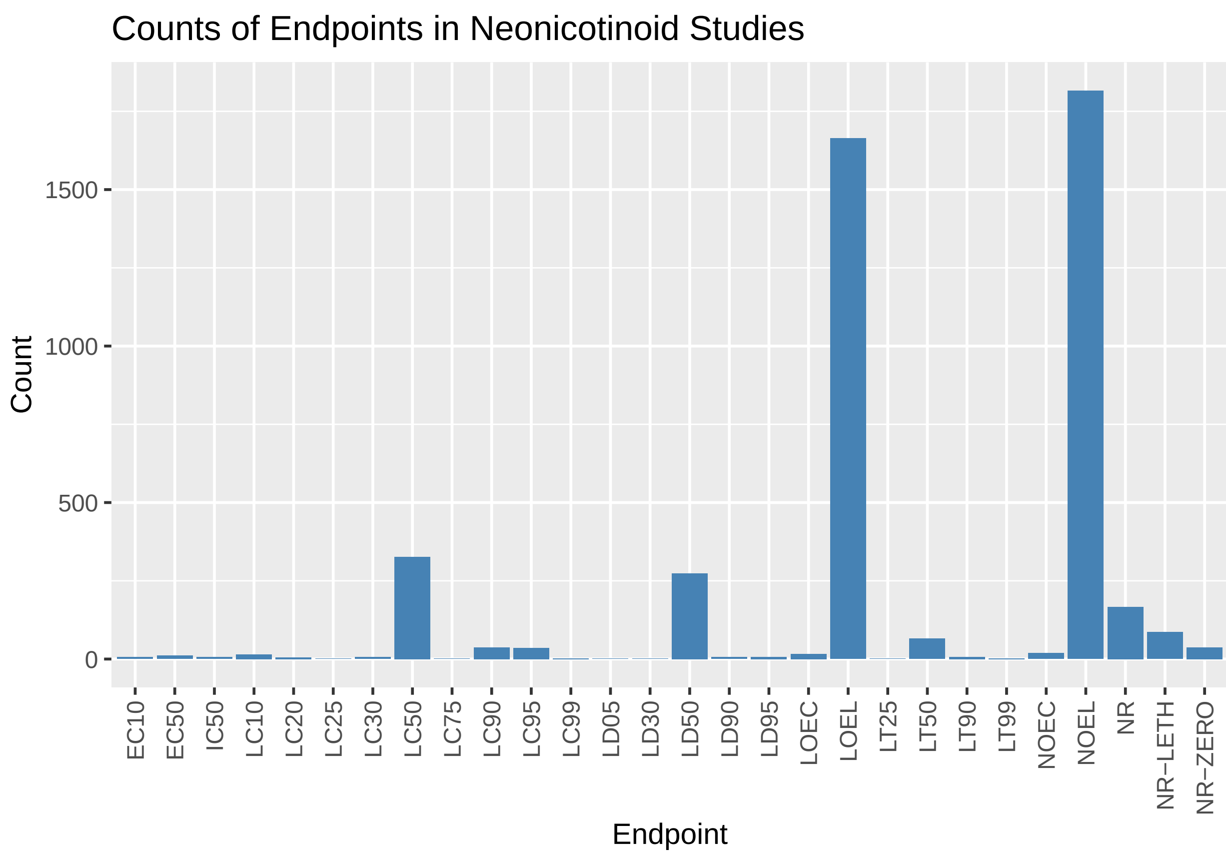
Interpret this graph. What are the most common test locations, and do they differ over time?

> Answer: The most common test location is lab, and it did change where the amount of studies conducted in lab increased largely after 2000.

11. Create a bar graph of Endpoint counts. What are the two most common end points, and how are they defined? Consult the ECOTOX_CodeAppendix for more information.

[**TIP**: Add `theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))` to the end of your plot command to rotate and align the X-axis labels…]

```
ggplot(Neonics, aes(x = Endpoint)) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "Counts of Endpoints in Neonicotinoid Studies",
    x = "Endpoint",
    y = "Count"
  ) +
  theme(legend.position = "top") +
  theme(axis.text.x = element_text(angle = 90, vjust = 0.5, hjust=1))
```



Answer: The two most common end points are NOEL and LOEL. NOEL is defined as "No-observable-effect-level: highest dose (concentration) producing effects not significantly different from responses of controls according to author's reported statistical test (NOEAL/NOEC)". LOEL is defined as "Lowest-observable-effect-level: lowest dose (concentration) producing effects that were significantly different (as reported by authors) from responses of controls (LOEAL/LOEC)".

Explore your data (Litter)

12. Determine the class of collectDate. Is it a date? If not, change to a date and confirm the new class of the variable. Using the `unique` function, determine which dates litter was sampled in August 2018.

```
class(Litter$collectDate)
```

```
## [1] "factor"
```

```
Litter$collectDate <- as.Date(Litter$collectDate)
class(Litter$collectDate)
```

```
## [1] "Date"
```

```r
unique_dates <- unique(Litter$collectDate[format(Litter$collectDate, "%Y-%m") == "2018-08"])
print(unique_dates)
```

```
## [1] "2018-08-02" "2018-08-30"
```

13. Using the **unique** function, list the different plotIDs sampled at Niwot Ridge. How is the information obtained from **unique** different from that obtained from **summary**?

```r
unique(Litter$plotID)
```

```
##  [1] NIWO_061 NIWO_064 NIWO_067 NIWO_040 NIWO_041 NIWO_063 NIWO_047 NIWO_051
##  [9] NIWO_058 NIWO_046 NIWO_062 NIWO_057
## 12 Levels: NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 ... NIWO_067
```

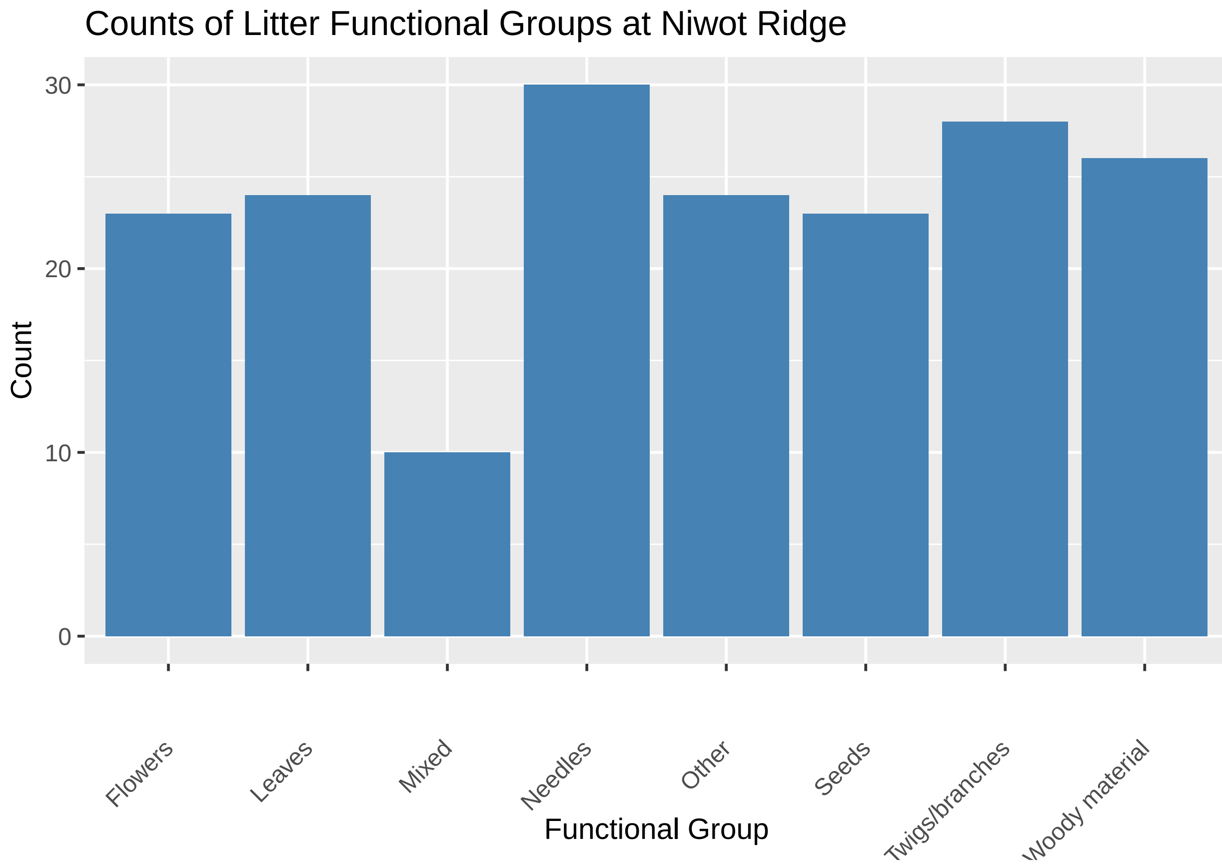
```r
summary(Litter$plotID)
```

```
## NIWO_040 NIWO_041 NIWO_046 NIWO_047 NIWO_051 NIWO_057 NIWO_058 NIWO_061
##       20       19       18       15       14        8       16       17
## NIWO_062 NIWO_063 NIWO_064 NIWO_067
##       14       14       16       17
```

Answer: Information obtained from 'unique' show the distinct values in the corresponding column and their level details, whereas 'summary' shows the frequency of each value.

14. Create a bar graph of functionalGroup counts. This shows you what type of litter is collected at the Niwot Ridge sites. Notice that litter types are fairly equally distributed across the Niwot Ridge sites.

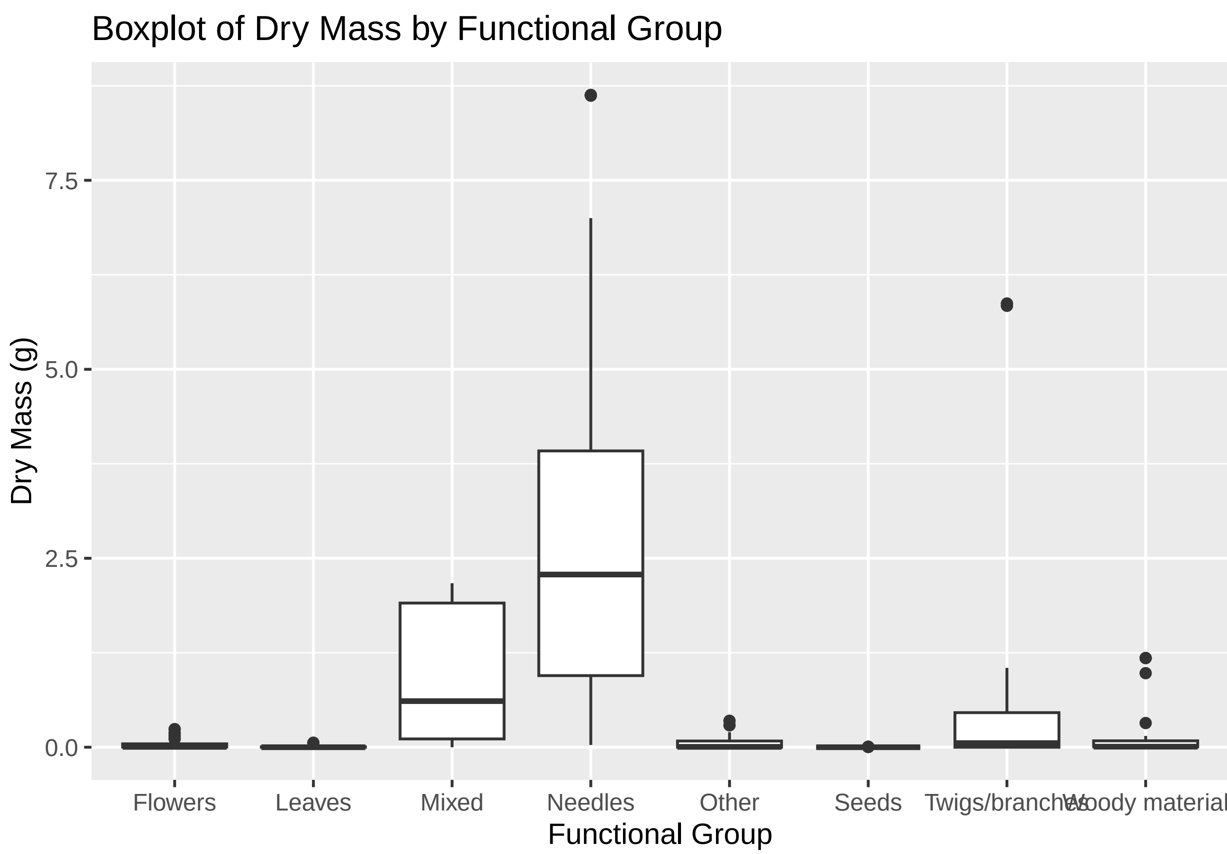
```r
ggplot(Litter, aes(x = functionalGroup)) +
  geom_bar(fill = "steelblue") +
  labs(
    title = "Counts of Litter Functional Groups at Niwot Ridge",
    x = "Functional Group",
    y = "Count"
  ) +
  theme(legend.position = "top") +
  theme(axis.text.x = element_text(angle = 45, vjust = 0.5, hjust = 1))
```



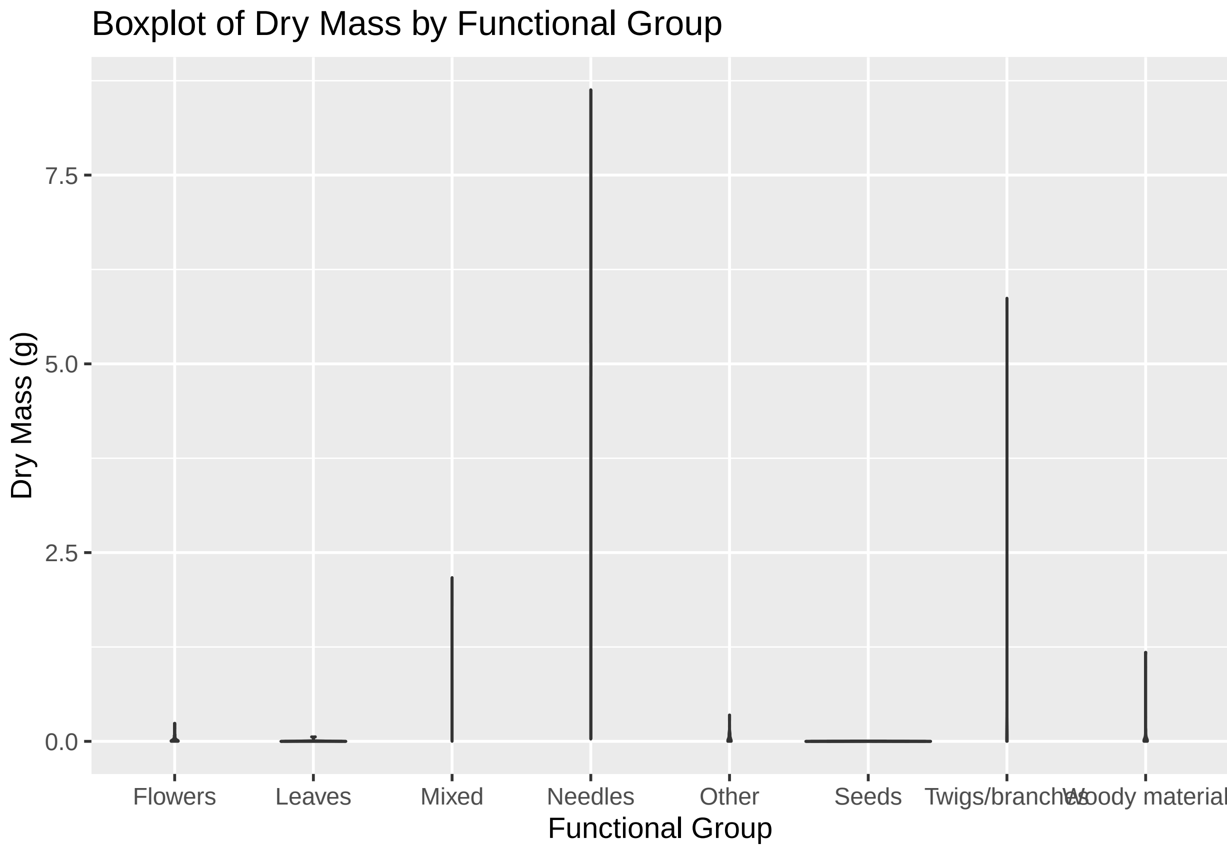
15. Using `geom_boxplot` and `geom_violin`, create a boxplot and a violin plot of dryMass by functionalGroup.

```
# Boxplot of dryMass by functionalGroup

ggplot(Litter) +
  geom_boxplot(aes(x = functionalGroup, y = dryMass )) + labs(
    title = "Boxplot of Dry Mass by Functional Group",
    x = "Functional Group",
    y = "Dry Mass (g)"
  ) +
  theme(legend.position = "top")
```




```r
# Violin plot of dryMass by functionalGroup
ggplot(Litter) +
  geom_violin(aes(x = functionalGroup, y = dryMass), draw_quantiles = c(0.25, 0.5, 0.75)) +
  labs(
    title = "Boxplot of Dry Mass by Functional Group",
    x = "Functional Group",
    y = "Dry Mass (g)"
  ) +
  theme(legend.position = "top")
```



Why is the boxplot a more effective visualization option than the violin plot in this case?

> Answer: The boxplot is more effective here because the dataset is not extremely large. When working with smaller sample sizes, violin would show misleading patterns thus being a less effective visualization option.

What type(s) of litter tend to have the highest biomass at these sites?

> Answer:Needles and twigs/branches tend to have the highest biomass.